A study on Human-Centered Emotion Classification based on Classical Machine Learning Models Capture in GoEmotions

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Abstract. As technology grows more "emotionally aware", models are expected to deliver a high-level of accuracy, while being extremely quick and easy to interpret. While Transformer-based models offer optimal performance, they complicate human-AI collaboration owing to its "black-box" nature and high computational cost. This study critically evaluates classical machine learning models - LinearSVC, Logistic Regression, and Random Forest against Transformer-based models such as RoBERTa and DistilBERT, on the GoEmotions dataset. While Transformer models establish a performance ceiling of 53% accuracy and 45% Macro F1-score, they present a "black-box" nature and significant computational overhead. On the other hand, classical models managed to achieve a peak accuracy of 53% and 43% Macro F1-score. Our research demonstrates that lightweight architectures can achieve comparable accuracy with a significant reduction in execution time, offering a more sustainable approach for real-time, human-centric affective interfaces.

Keywords: HCI, Green AI, Explainable-AI, ML, Transformers.

1 Introduction

Machines being able to understand and respond to human emotions has gone from being a research pursuit to an essential requirement in contemporary technological systems. Historically, this field was pioneered with the concept of "Affective Computing". It suggests that in order to create systems that are truly intelligent in the way that humans are, computers must be capable of interpreting human emotions. This transformed Natural Language Processing (NLP) from simplistic sentiment analysis, binary classification of "positive" or "negative" to developing systems that can detect the intricacies of vast human emotions. These emotions are based on well-established frameworks such as the Ekman Taxonomy of universal basic emotions.

While deep learning models and Transformer-based architectures are considered the standard for accurate emotion recognition. These "black-box" models have extremely low interpretability due to their nature and have major trade-offs in terms of computational cost and latency, resulting in reduced performance efficiency. The high energy and extensive hardware requirements of deep learning models hampers Human-Computer Interaction (HCI). Thus, offering limited insight to the human collaborator about the specific linguistic triggers that formulated the model's decision. By focusing on interpretable machine learning architectures, this research follows the prevalent paradigm of Explainable AI (XAI). Ensuring that the classifications are not just accurate but also verifiable and justified. Additionally, in real-time HCI environments, for example, mental health support tools, system latency can become a significant barrier in the natural progression of the interaction. By highlighting the speed of a model to be vital in the success of such systems.

This study investigates Human-Centered Emotion Classification by utilizing GoEmotions dataset - a vast corpus of 58,000 Reddit comments. To evaluate the ability of both classical machine learning models and transformer-based models in recognising the complexities of human sentiment. The primary objective of this study is to bridge the gap between high-dimensional data processing and interpretable HCI. The human-centeredness of the study stems from the use of Classical Machine Learning models and how they offer greater interpretability when juxtaposed with the "black-box" nature of deep learning models. Unlike broader sentiment analysis, our approach utilizes the 27 distinct emotion categories of GoEmotions dataset to further categorize in 7 emotion categories according to Ekman taxonomy. To explore the granularity of human expression, from joy to disgust.

In order to precisely define the purpose of this study, we must highlight what differentiates our approach from that followed in existing research papers. Contemporary studies often exclusively focus on achieving maximum accuracy using Transformer Models like BERT and BiLSTM. While our study prioritizes computation efficiency, in order to strike a balance between performance and efficiency. Moreover, while a majority of the earlier studies work with "clean" data such as the ISEAR dataset, our study aims to tackle the inherent "noise" of natural language, inclusive of slang, sarcasm and modern-day diction, as found in GoEmotions dataset. Finally, unlike previous studies that focus purely on algorithmic tuning and deep learning architectures operating on a black-box model, our work is "human-centered" and takes an HCI-First perspective. Prioritizing interpretability for the human collaborator, allowed by models such as Logistic Regression and LinearSVC.

To establish a baseline for performance, this study looks at two Transformer-based models, RoBERTa [25] and DistilBERT [26], to quantify the specific trade-offs between metrics like accuracy and F1-score, and system latency when compared to classical models.

Ultimately, this study examines the accuracy and computational performance of classical machine learning models when compared to deep learning models that benchmark a baseline of 53% accuracy and 43% Macro F1-score while transformers could only achieve 52% accuracy and 45% Macro F1-score. This paper demonstrates how

lightweight machine learning architectures can contribute to more responsive and transparent affective computing interfaces.

2 Literature Review

We analysed many of the papers which claimed to be using machine learning for sentimental analysis. Bharti et al.[1] were using hybrid models such as classical machine learning and Deep learning together. Which loses the aim of creating a transparent or white box model by creating a mix of them. While using highly lab-curated dataset such as ISEAR dataset (7,600) [2],WASSA (4334) [4] or Emotion-Stimulus (2500) [6]. These Datasets tend to give us comparatively higher accuracy but due to its smaller size compared to GoEmotion dataset (58,000) [3] are more prone to overfitting models and losing the concept of organic raw emotions. Also the number of emotions are mismatched in these dataset which do not give us any clear idea of our model as a whole. By integrating Deep Learning in classical machine learning the computational cost and latency also faced a spike. As their[1] classical model already has a bench mark of 78.97% accuracy which is just increased to 80.11% by using hybrid model just increasing by only 1.14% which is obviously as over consumption of carbon.

The work by Pratama et al.[8] and et al.Rasyad [9] sharing the same dataset as well as co-author. They are very specifically focused on Indonesian text dataset which is extracted from Indonesian language tweets through web-crawling. They did very thorough formal preprocessing which can lead the model to lose raw emotions from the text. While one of the papers used modern techniques like Word2vec [9] and performed poorly. They didn't explore what could be the reason for the poor performance. While using techniques such as TF-IDF [10] and BoW [11] in most of their experimentation they could use bigram or n-gram to capture a more meaningful insight from tweets.

While Shrivastava et al.[12] attained high accuracy with their ISEAR dataset [2] it is a waste of computational resources as models such as Bi-LSTMs [17] need high gpu and are data hungry. Such small dataset causes bottlenecks for heavy models like these and are prone to overfitting.The ISEAR[2] dataset used by Shrivastava [12] and Bharti [1] was taken in the 1990s which is somewhat irrelevant to today's slangs, failing to fulfill today's needs. The model created by Pratama et al. [7] and Rasyad et al. [8] would work fine on this dataset [2] if hypertuned better, creating less carbon footprint and higher latency. Saji's [13] work also depicts the same pattern of Bharti et al. [1] for an increase of 4% from Logistic regression [14] and BoW [11] giving 88% accuracy while LSTM [16] jumping to 92% accuracy to BERT [15] giving 91% accuracy. For chasing those 4% accuracy researchers keep going to deep learning and transformers neglecting the computational cost, carbon footprint and latency delay. Is crossing that 4% plateau even worth that cost?

A research conducted by Ibnath et al. [20] evaluates traditional machine learning models such as SVM and Logistic Regression in comparison to one deep learning model (BiLSTM) for emotion detection. The authors carried out a comparison between three feature extraction techniques: TF-IDF, Count Vectorization and N-grams. The study saw an accuracy of 88.09% [20] being achieved using N-grams with Logistic

Regression when applied to a dataset consisting of 16000 data points divided into 6 basic categories. However, this study exclusively focuses on accuracy and F1-score, and does not take into consideration computational cost and time efficiency, which are critical factors in real-world human-focused applications.

On the other hand, a research paper by Zghoul et al. [21] carries out a comparison between 3 classical machine learning models using a balanced dataset of 106,000 sentences across 11 emotional categories, utilising TF-IDF with trigram approach. While this research advocates for these models due their low computational cost and transparency as compared to deep learning architectures, it does not address the "human-centeredness" aspect, and how these models perform on real-world, messy data when compared to their balanced, curated dataset. Our study on the other hand, uses the GoEmotions dataset, rife with the linguistic complexities of the natural language, using 58,000 user comments from Reddit.

We also explored a study by Suryaningrum et al. [22], a research comparing TF-IDF and Count Vectorizer using Support Vector Machine (SVM) to identify hate speech within Indonesian Twitter data. Using a large dataset of 100,000 tweets, the study concludes that TF-IDF is superior, providing an accuracy of 88.77% [22]. However, this paper only delves into a binary classification task, of categorising Hate Speech and Non-Hate. Our study delves into a more complex classification exercise by classifying data from the GoEmotions dataset into 7 distinct emotional categories.

A study published in October 2024 by Saif et al. [23] compared classical models - Logistic Regression, Extra Randomized Tree, Voting Classifier, SGD and LinearSVC - using TF-IDF, Bag-of-Words, and N-grams as feature extraction techniques. They tested the performance of these models on the ISEAR and AIT-2018 datasets. Their research concluded that LinearSVC with N-grams achieved the highest accuracy score, of 88.63% [23]. While this study successfully evaluates the models with regards to emotion detection, it lacks the "human-centred" approach required for building real-time systems where latency and transparency become essential factors.

Adewumi et al. [24] in their study utilized Transformer-based NLP for text, CNN-RNN for speech and ResNet for facial expressions. The study achieved a high multimodal fusion accuracy of 92.3% [24]. This study relies heavily on heavy architectures, models that require significant computational power, making them difficult to deploy on low-power devices. On the other hand, our study shifts focus towards lightweight architectures, and argues that classical machine learning models can offer a comparable accuracy while significantly increasing computational efficiency.

3 Methodology

3.1 Dataset Description and Taxonomy Mapping

This study utilizes the GoEmotions dataset, a human-annotated corpus of 58,000 Reddit comments. Originally containing 27 fine-grained emotion categories, the dataset reflects the complexity of natural language. To ensure biological relevance and reduce data sparsity, we mapped these 27 labels into the Ekman Taxonomy, which consists of six universal emotions: Anger, Disgust, Fear, Joy, Sadness, and Surprise, alongside a

Neutral category. This mapping allows the study to bridge the gap between complex social media expression and fundamental psychological frameworks. The share of each emotional category in GoEmotions dataset is as shown in Fig.1.

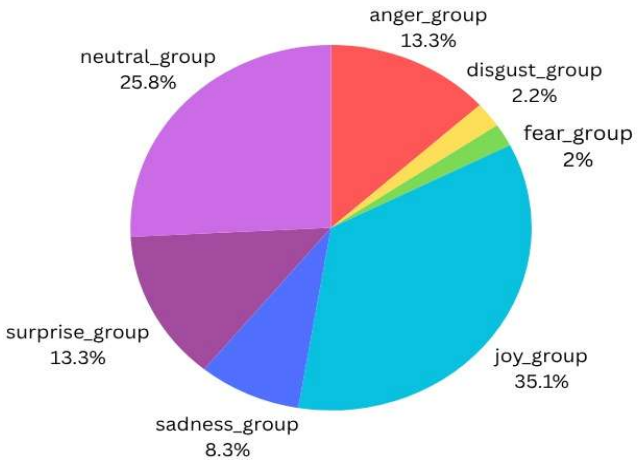


Fig. 1. Share of each emotional category in GoEmotions dataset

3.2 Data Preprocessing

To address the "noise" inherent in Reddit data—such as slang, modern diction, and inconsistent casing - a multi-stage preprocessing pipeline was implemented:

- Reduced Redundancy: These 57,000 comments were reviewed by 3 to 5 different humans for different opinions; after gathering all opinions and storing them in one place the dataset became a collection of 211,000. So, we first removed the redundant value to reduce the dataset to 58,000.
- Removing extra data : There were unnecessary data columns like ['id','author', 'subreddit' ,'link_id', 'parent_id', 'created_utc', 'rater_id'], which had nothing to do with our research.We also removed the column which were marked as unclear exemplar or were empty.
- Ekman Mapping: We classified 27 emotions into 6 universal emotions with neutral for better classification results.
- Emotion Prioritizing: We only took 1 emotion per comment and tried that we first skip through neutral to focus on core emotions and reduce class imbalance.
- Lemmatization: Reducing words to their dictionary base forms to ensure the models recognize semantic similarities despite morphological variations. This step of preprocessing was bypassed for Transformer-based models.
- Test - Train split: The most common step for any machine learning or deep learning process to split data for model training and testing we took 80% of our data in training and 20% in testing.
- Vectorizing: The bridge between text and machine language. We used both CountVectorizer and TD-IDF Vectorizer to see the difference in result. Since, deep

learning models utilize pre-trained high-dimensional embeddings, there is no requirement for manual feature extraction.

3.3 Model Selection and Architectural Overview

In alignment with the Explainable AI (XAI) and Green AI goals, we selected three classical machine learning models known for their mathematical transparency and low computational latency in this study and two Transformer-based models known for their computational efficiency and high accuracy:

First we can divide the classical models into 2 groups:

Vectorizers. These are the brain behind converting our text data into numerical data to feed into our machine learning models. As we know machines can't understand english or any other language it only understands numerical data so it's essential that we convert our textual data into numerical data. For the process of conversion we used two different models as discussed below.

Countvectorizer. This model for conversion to textual data in numerical data is based on the concept of bag of words (BOWs) theoretical concepts. In this we count the occurrence of words and model understands that more the number of words more importance that word holds

TF-IDF Vectorizer (Term Frequency-Inverse Document Frequency). This model for conversion of textual data into numerical data works a bit different from CountVectorizer as it just not counts the occurrence of words it weighs frequency against rarity, it penalizes the common words that appear in almost every sentence (eg. 'A', 'An','The'). Further assigning weight to words according to their relevance in the classification.

1. Term Frequency (TF): This represents the relative frequency of a term t within a specific document d.

$$TF(t,d) = \frac{f_{t,d}}{\sum_k f_{k,d}} \tag{1}$$

where $f_{t,d}$ is the number of time t appears in in the document d and $\sum_k f_{k,d}$ is the total number of terms in document d.

2. Inverse Document Frequency (IDF): This measures how much information the word provides—whether it is common or rare across all documents.

$$IDF(t,D) = \log\left(\frac{N}{|\{d \in D: t \in d\}|}\right) \tag{2}$$

where N is the total number of documents in the corpus D and $|\{d \in D: t \in d\}|$ is the number of documents where the term t appears.

3. *TF-IDF Score.* The final weight is the product of the TF and IDF scores.

Bag of Words (BOW). It is a concept used by CountVectorizer. It loses the context and word order or any grammatical relationship in a sentence. It just focuses on the

frequency of a word not taking accountability of its effect on outcome. It's aimed to reduce computational complexity by reducing mathematical operations.

N-Grams. It's a continuous sequence of N numbers of items, typically words of characters. It's used to somewhat preserve the context of syntactic structure and context within a vectorizer. There are multiple names of N-Grams such as Uni-Gram (n=1), Bi-Gram (n=2), Tri-Gram (n=3),Four-Grams(n=4) etc. It helps us increase context awareness as well as our accuracy, F1-score etc.

3.4 Model Architectures

Linear Support Vector Classifier (LinearSVC). Linear support vector classifier (LinearSVC) is a discriminative supervised learning algorithm that finds the optimal hyperplane that separates multiple classes. Unlike traditional SVMs that may use non-linear kernels like RBF or polynomials, it uses linear kernels. It's specially optimized for linear kernels making it more efficient to work with CountVectorizer, N-Grams. Its main aim is to increase the gap between hyperplane and support vectors.

Logistic Regression. Logistic Regression is a robust classification model that works with one or more independent variables to find dependent outcome variables. Unlike linear regression which gives continuous values, it uses sigmoid function to make sure all values lie between 0 and 1. The classification is based on a certain threshold value (mostly 0.5) which determines the dependent variable falls in which class. The Sigmoid Activation Function:

$$\sigma(\mathrm{z}) = \frac{1}{1+e^{-z}} \tag{3}$$

where $\sigma(\mathrm{z})$: Represents the output probability value, constrained between 0 and 1 and z represents the logit function:

$$z = w_1 x_1 \; + \; w_2 x_2 + \ldots + w_n x_n \; + \; b \tag{4}$$

w_n represents the weights, parameters the model learns to assign importance to specific features.

x_n Represents the Input Features and b represents the Bias term, a constant allowing model to shift the decision boundary for a better fit.

Euler's number represented by e is a mathematical constant (2.718)

Random Forest. Random Forest is an ensemble classification algorithm that works on multiple decision trees during training to improve predictive accuracy and reduce the chances of occurrence of overfitting. Each tree is trained on random features and dataset. Unlike decision trees it implies randomness and bagging. For final classification the average of majority votes of individual trees is taken into account.

DistilBERT. DistilBERT is a fast and light Transformer model based on the BERT architecture. It offers 97% of BERT's performance while being 40% smaller and 60%

faster, making it an ideal "middle-ground" for comparison between deep learning architectures and classical ML models. This model was developed by Hugging Face and utilizes approximately 82 million parameters.

RoBERTa. RoBERTa is a robust, optimized BERT model, that utilizes larger batches and more data. It serves as our high-accuracy point of comparison to test the upper performance limits for a model when applied to the GoEmotions dataset under the Ekman taxonomy. This model was developed by Meta AI Labs and operates with approximately 125 million parameters.

3.5 Model Evaluation

Accuracy. For classification accuracy is considered as one of the best metrics to evaluate the performance of emotion classification models. While it's considered as one of the best metrics it can miss leading to highly imbalanced data as ours where joy takes 35% where fear takes 2% of our dataset. If our model crams the frequently occurring classes ignoring classes with less frequency which can mislead our accuracy metrics. Accuracy can be computed using the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP+TN+FP+FN} \tag{5}$$

where TP (True Positive) are the correctly predicted positive instances, TN (True Negative) are the correctly predicted negative instances, FP (False Positive) are incorrectly predicted positive instances, FN (False Negative) are incorrectly predicted negative instances.

F1-score. F1-score is the harmonic mean of precision and recall providing the best metric to evaluate classifier models particularly in the base of highly imbalanced data as ours. While accuracy can be misleading in highly imbalanced dataset F1-score has both the ability to identify both the positive instances (recall) and the ability to find that the identified positives are true correctly (precision). By using harmonic mean instead of simple arithmetic mean it penalizes the extreme values. F1-score can be computed using the formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{6}$$

Where:

$$Precision = \frac{TP}{TP+FP} \tag{7}$$

$$Recall = \frac{TP}{TP+F} \tag{8}$$

3.6 Tools

Kaggle. To ensure that the proposed model can withstand computational reproducibility we ran the proposed model within a standardized Kaggle environment. It helps us

provide stable computational time and cost, making it easier to estimate the computational cost for different devices. Kaggle provides us with a fixed resources allocations (e.g., RAM, CPU, GPU, TPU).

Python 3.12.12. The proposed model utilizes Python (v3.12.12) as the core implementation language for the entire proposed model due to its friendliness in the field of machine learning and human readable syntax making it reliable for developing transparent yet maintainable codes**.**

SpaCy. We implemented SpaCy as one of our many pre-processing steps on the GoEmotion dataset. It is a Python framework used to implement a few of NLP steps. We specifically performed Lemmitization, which helps to reduce inflected word forms to their linguistically accurate root.

Scikit-Learn. Scikit-Learn aka Sklearn served as the core computational engine for the proposed model as it helped us in every step of our journey from splitting data in test and train to vectorizing to fitting in different models giving us different accuracy, macro F1-score. The library functioned as our primary driver for our experimental results

Pandas. Pandas served as a data manipulation backbone providing a large variety of data manipulation and cleaning techniques with ease. It seamlessly helped with data cleaning and preprocessing. It reduces our work by converting data in data structure called DataFrames which have specialized well defined data manipulating tools and features.

Matplotlib. We explicitly used Matplotlib.pyplot to create visuals for the proposed model, while matplotlib provides us with an object-oriented visualization library. We utilized the matplotlib.pyplot module to manage the plotting state and visualization. It automates the creation of figures and axes reducing our load of stating it for each visual.

Transformers. The Hugging Face Transformers library was used to implement the DistilBERT and RoBERTa models, providing access to pre-trained weights and optimized tokenizers.

PyTorch. This Python library served as the underlying deep learning framework for the deep learning models, facilitating GPU-accelerated training and loss calculation. It aided in the iterative fine-tuning of the high-parameter Transformer architectures on the GoEmotions dataset.

4 Result

The major focus of this study was to see how traditional machine learning models are more efficient, interpretable and more computationally efficient than the transformers and deep learning models. The results are quite satisfactory for the proposed study as you can see.

Table 1. Comparative analysis of performance metrics and computational latency across all evaluated models.

Models	Macro F1 Score	Accuracy	Time Taken (in seconds)
Logistic Regression (TF-IDF)	43	51	50
Logistic Regression (CountVectorizer)	43	53	85
LinearSVC (TF-IDF)	42	51	5
LinearSVC (CountVectorizer)	39	50	15
Random Forest (TF-IDF)	35	52	555
Random Forest (CountVectorizer)	36	53	605
DistilBERT	44	52	303
RoBERTa	45	52	605

In Table 1 we can see that even if Accuracy is almost equal for all of the given models and vectorizers the Macro F1-Score varies up to 10 points where accuracy only 3 points it shows that some of our models such as Random Forest with TF-IDF and CountVectorizer have lowest Macro F1-Score out of all still reaching highest accuracy proves that our model is predicting most occurring class again and again to increase the accuracy of model, thus ignoring less occurring classes. But our model with higher Macro F1-Score and Accuracy tries to focus on minority classes too. Specifically, the RoBERTa and DistilBERT benchmarks achieve the highest granularity in emotion detection, effectively bridging the gap where classical models struggle with data sparsity. This difference can also be seen in the Confusion Matrices of Random Forest (Fig. 3) and Logistic Regression (Fig. 2), as well as the integrated RoBERTa (Fig. 4) and DistilBERT (Fig. 5) models."

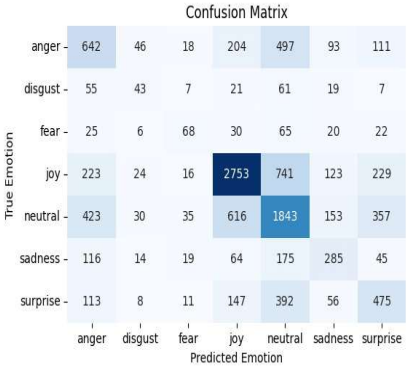


Fig. 2. Logistic Regression (CountVectorizer)

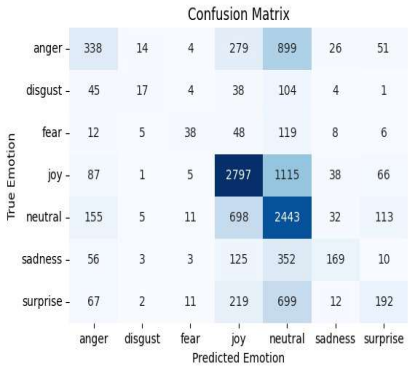


Fig. 3. Random Forest (CountVectorizer)

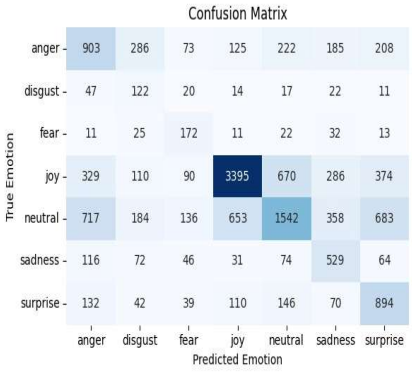


Fig. 4. RoBERTa

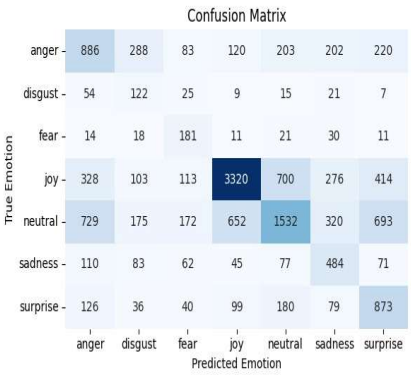


Fig 5. DistilBERT

In Fig. 3 we can see how much Random Forest is concentrated in joy and neutral blocks which are the dominant classes of our dataset. Meanwhile in Fig. 2 Logistic Regressor is comparatively less concentrated on dominant blocks i.e. joy and neutral and a bit more focused on other classes also doubling the accuracy of minor classes.

This trend reaches its peak with the Transformer benchmarks; as seen in Fig. 4 (RoBERTa) and Fig. 5 (DistilBERT), the models demonstrate a significantly sharper diagonal, particularly in sparse categories such as 'Anger', 'Fear', and 'Surprise'.

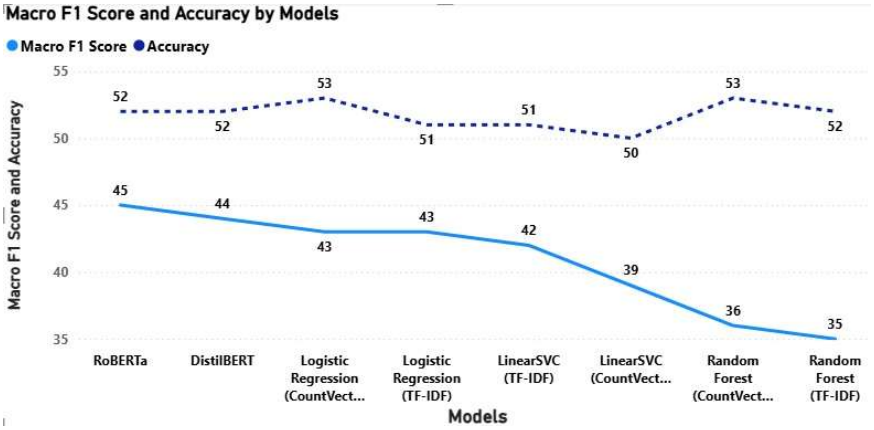


Fig. 6. Comparison of Macro F1-Score and Accuracy across the studied models

Fig. 6 compares Macro F1-Score and Accuracy across all of our models, revealing a negligible performance gap between classical and Transformer-based. It is clearly visible that even with the Deep Learning models, produce figures that lie within the 2% range, highlighting that deep learning models do not offer a significant predictive advantage over classical models in this context. As we also focus on other aspects like

XAI, these numbers help us how AI understands and works on data instead of "Black-box" model.

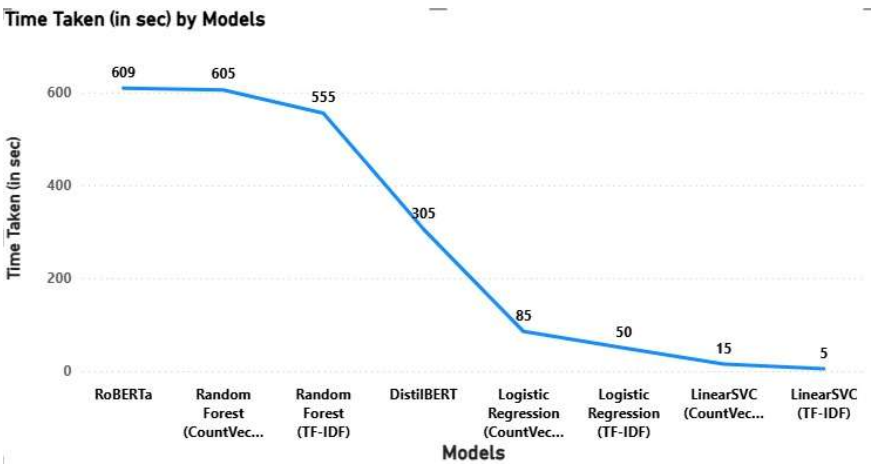


Fig. 7. Comparison of Execution Time across the studied models

Fig. 7 depicts the time taken by our different models, highlighting the significant jump in latency required for Deep Learning. As shown in the figure, the LinearSVC (5s) and Logistic Regression (50s) models demonstrate a massive efficiency advantage over RoBERTa (605s) and DistilBERT (303s). These numbers prove that the Transformer benchmarks require up to 120 times more processing time for a marginal 2% gain in performance.

This is one of the most important aspects of the proposed study as it not solely focuses on metrics such as Accuracy and Macro F1-Score but carbon emission, computational efficiency and green AI too. These numbers help us to choose the most optimal model with least amount of trade-off with metrics such as Accuracy, Macro F1-Score.

5 Conclusion

As we proceed to conclude the proposed study there are multiple aspects to focus on to decide which architecture is most optimal to use in real world AI solution and Human centric systems.

In terms of Accuracy alone, all evaluated models - including the 125-million parameter RoBERTa (53%) and the 82-million parameter DistilBERT (52%) fell within a marginal 2-3% range of the classical architectures. This suggests a performance plateau, where an increase in architectural complexity yields negligible gains in classification success.

But as we know Accuracy can easily be manipulated by model by cramming dominant classes.

Here comes the Macro F1-Score which tells us beyond accuracy, how well our model performed on all classes dominant or minor. While RoBERTa established the performance ceiling with a 45% Macro F1-score, it represents only a 2% improvement over Logistic Regression (43%).

Within the classical models, as discussed above Random Forest proved to be cramming dominant classes whereas Logistic Regression outperforms other models with both vectors CountVectorizer and TF-IDF.

When we talk beyond model efficiency and go toward computational efficiency which is also one of the important goals of the proposed study. For the aspect of computational efficiency, time taken by each model is one of the most important factors as our models ran on Kaggle, each model ran in a controlled environment.

RoBERTa (605s) and DistilBERT (303s) required substantial processing time, whereas LinearSVC (5s) delivered near-identical results 120 times faster.

From this point of view LinearSVC is by far the clear winner. By merely taking 5 to 15 seconds and giving Accuracy of 51 and Macro F1-Score of 41 LinearSVC is the clear winner for computational efficiency.

Taking all the values in account LinearSVC provides the best trade-off between time taken and model evaluation metrics.

6 Future Scope

This study establishes a comparative baseline for both classical and transformer-based models in human-centred emotion classification. However, the research topic could be further explored by addressing other aspects that may enhance performance without compromising the transparency and low latency required for human-first applications.

Moreover, in order to further substantiate the study on Machine Learning models in human-first contexts, the proposed LinearSVC and Logistic Regression models should be integrated into live applications, for example, mental health chatbots. Conducting user studies would provide qualitative data on the effectiveness of XAI in these systems.

Additionally, while this study looks at “noisy” Reddit comments, it could be expanded further to look at other noisy environments such as short-form video transcripts and Discord logs. This would build upon the exploration of classical model performance with respect to varying modern linguistic patterns.

Finally, future research should aim to quantify the “Green AI” aspect, comparing the carbon footprint of training and deploying these lightweight machine learning architectures, with Transformers, further evaluating the performance-efficiency trade-off.

Below are the real - world use cases where the proposed study can be implemented:

- Responsive Human-Computer Interaction (HCI) Interfaces: Any real-time system that is designed to read and react to human emotions can benefit from the proposed study. By using lightweight classical models, systems can process emotional states instantly without the unnecessary delay caused by Transformer models.
- Deployment on Edge Devices: The proposed study criticizes deep learning architectures (like Transformers) for requiring high GPU power. The classical models

evaluated (which take only 5 to 50 seconds to train) can be easily deployed locally on mobile phones, wearables, or older hardware without requiring expensive computing resources.

- "Green AI" and Sustainable Enterprise Processing: For companies dealing with massive amounts of customer feedback or social media data, running deep learning models can be expensive to run and a massive carbon footprint to deal with for merely a 4-5% increase in accuracy. The proposed study helps enterprises to perform large-scale sentiment and emotion analysis efficiently, heavily reducing financial costs and carbon emissions.
- Emotion Tracking in Digital Communities: The models were specifically tested on the GoEmotions dataset (Reddit comments) to handle slang, sarcasm, and modern diction. The paper suggests these lightweight models are highly applicable for monitoring unethical, modern, messy text environments like Discord logs, Gaming chat rooms, Reddit, messaging apps etc to gauge community sentiment or detect toxic behavior quickly.

In short, the practical application of the proposed study is building fast, transparent, and eco-friendly Human-Computer Interaction systems for real-world environments where computing resources are limited and explainability is mandatory.

7 References

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